

imx6 multiple ethernet drivers - startup configuration concept

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Run init twice

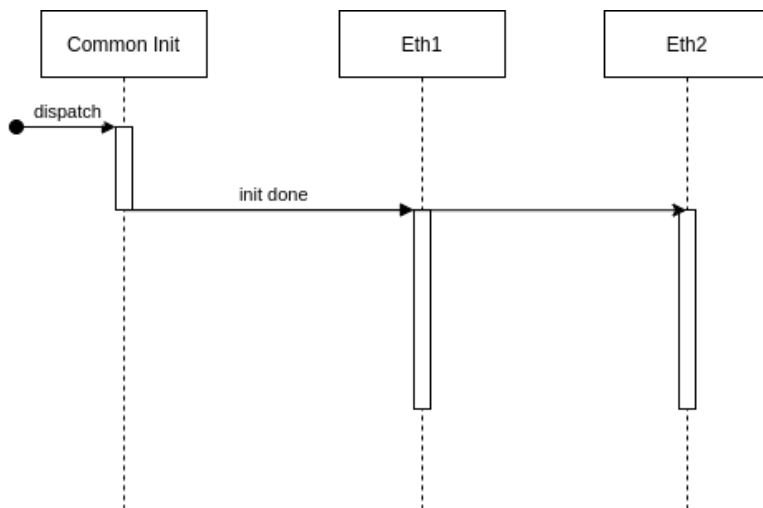
- creates synchronization issues

Master / slave components

- doesn't scale
- using the slave port requires the master one to be active

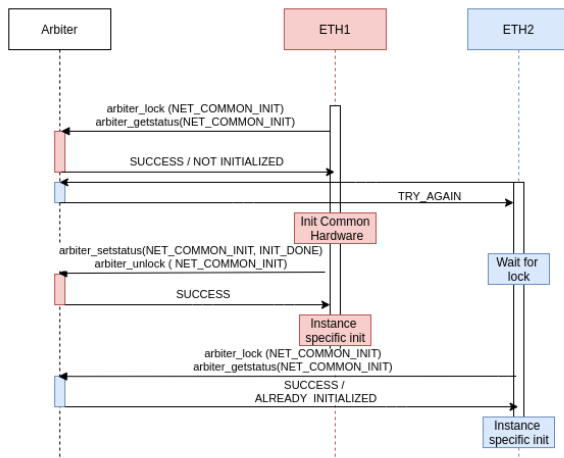
Global init component

- one component doing all the HW initialization, the others waiting for an init_done event (1 to many events are not possible)
- component's configuration would be stored in another component
 - IO Addresses and GPIO pins used
 - MAC addresses
- the init component needs to know all HW specific information for all components it's initializing
- all drivers need to be adapted / split in 2



Arbiter component

- rpc interface over a mutex (semaphore because we need trywait not to block in the RPC) + history allowing it to be conditionally used
- no HW specific code in this component



Blocks/features that need protection get a distinguishing name (ex. HW_GPIO_COMMON_INIT, HW_ETH_COMMON_INIT)

The arbiter provides an RPC interface allowing us to:

```

arbiter_lock(HW_ETH_COMMON_INIT)
arbiter_unlock(HW_ETH_COMMON_INIT)
arbiter_getstatus(HW_ETH_COMMON_INIT)
arbiter_setstatus(HW_ETH_COMMON_INIT, status)
  
```